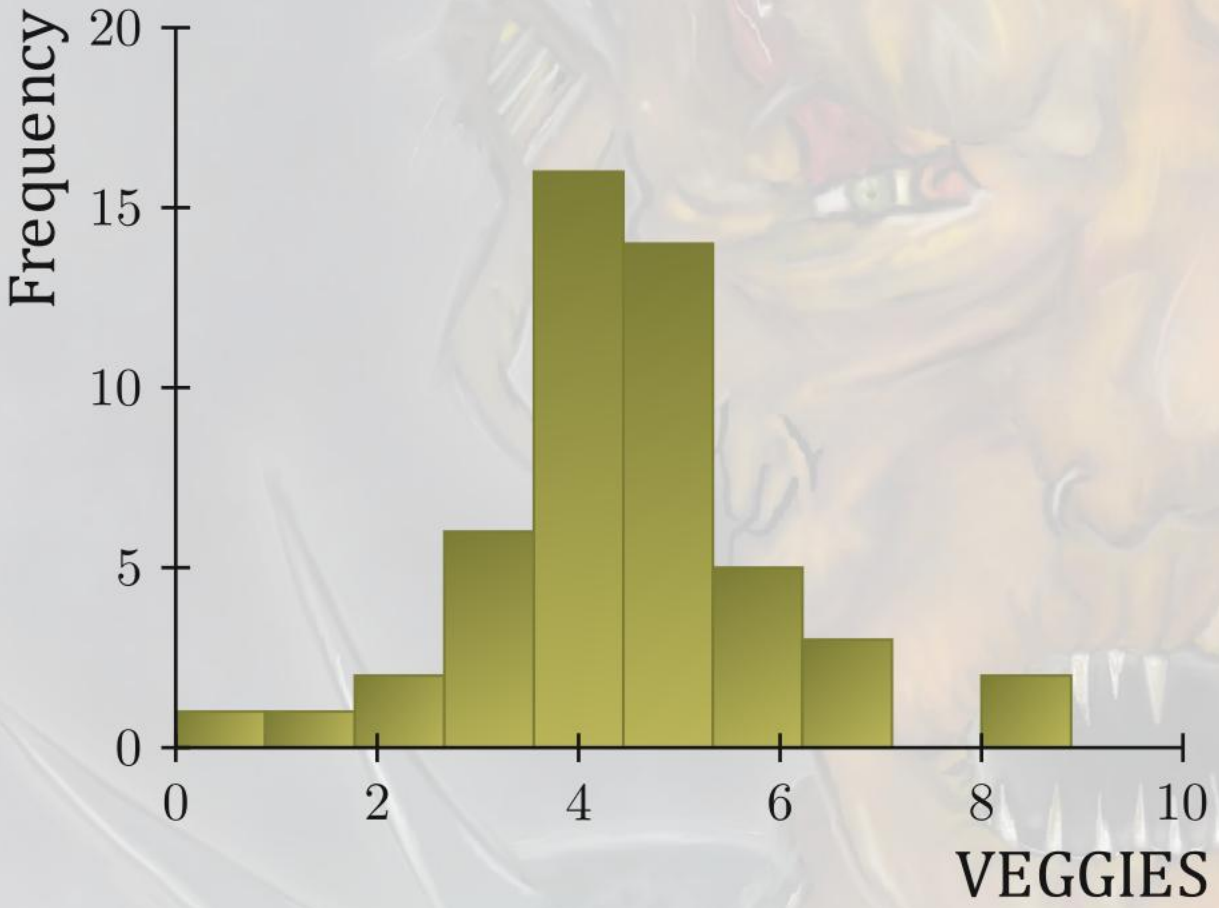
Statistics: week 1

Variables and charts

Orcs have an unhealthy and one-sided diet

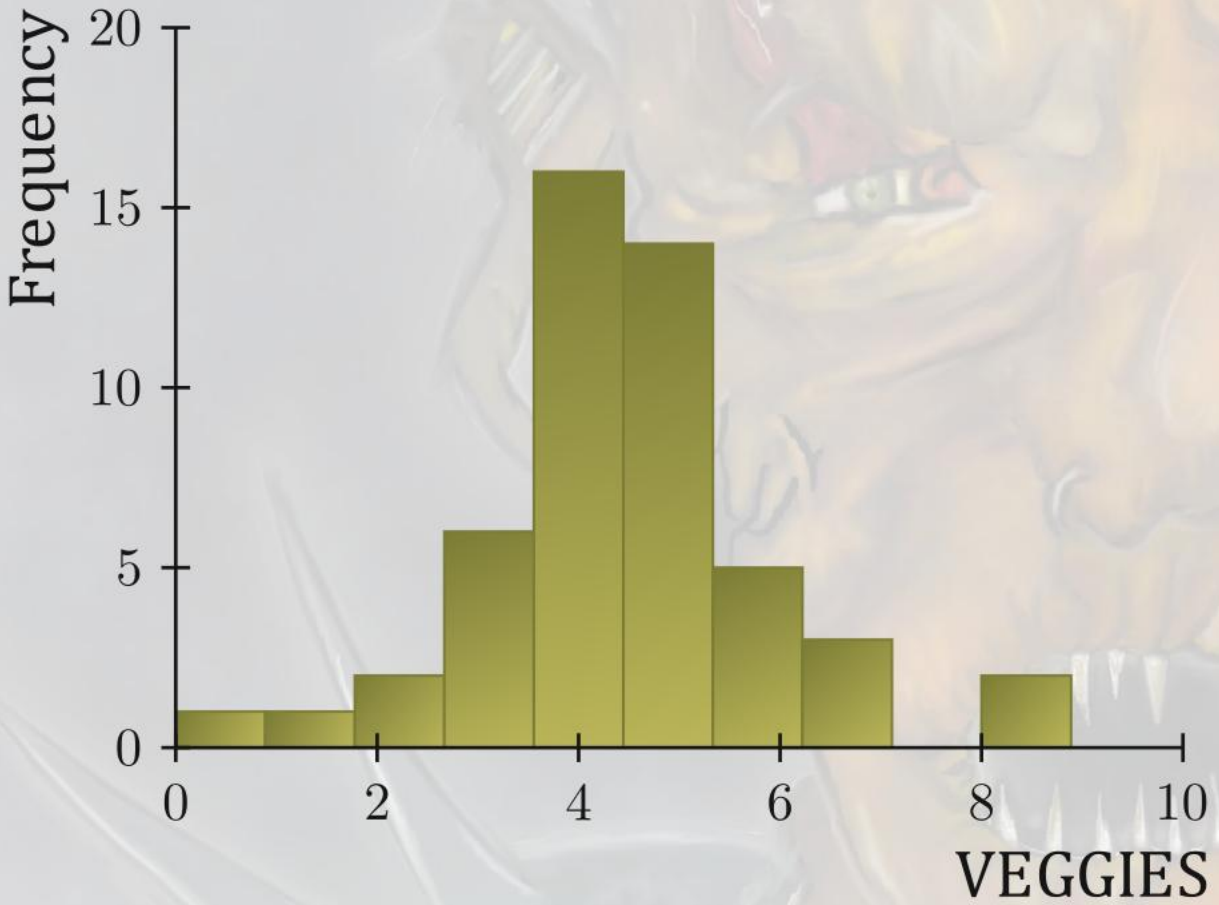


What is this graph?

- Latest order from the Dark Lord: 'Henceforth, you shall eat 1 handful of vegetables every day (or be skinned alive)!'
- The Lord immediately spies on a number of his minions for a week, to see if they obey him
- To understand this study, we must identify:

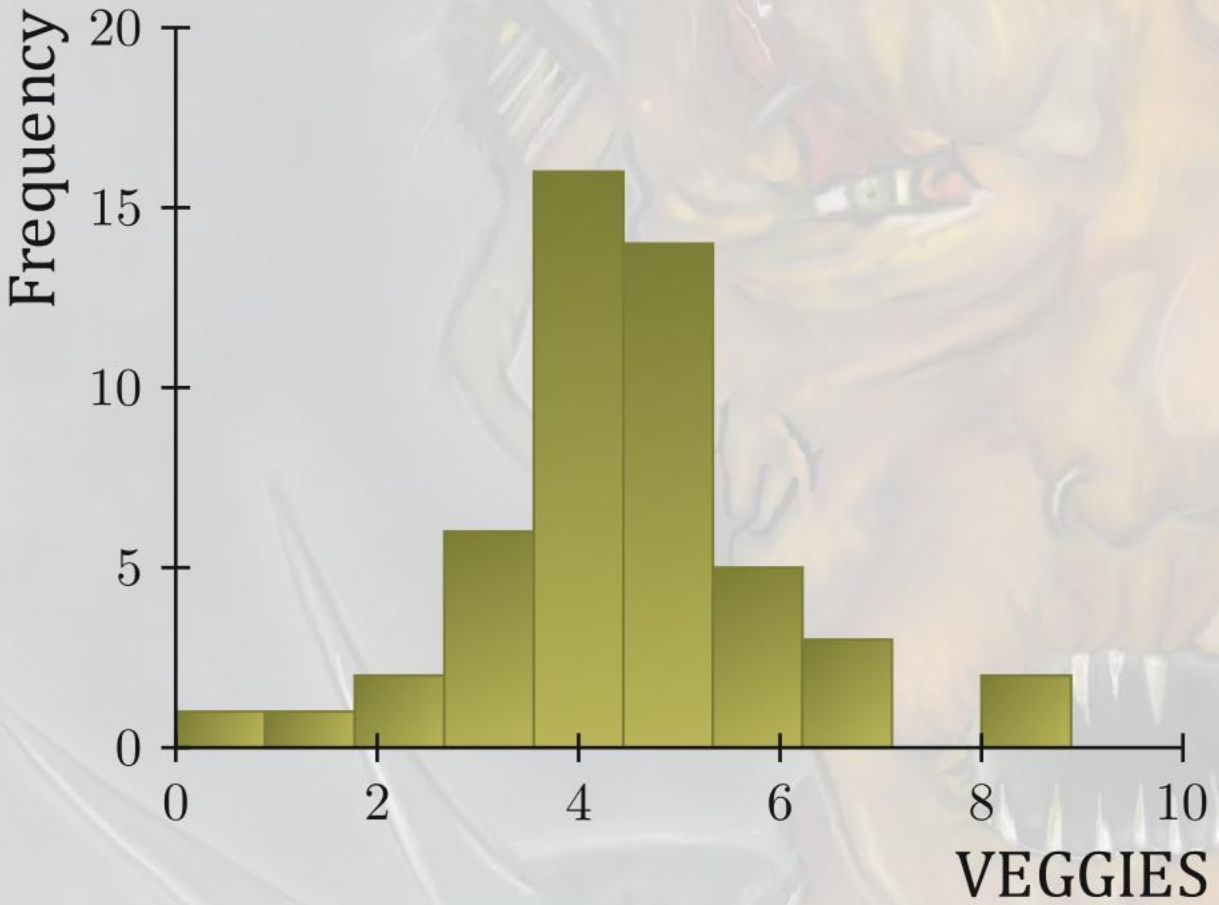
1. The **observational unit**
2. The **number of observations**
3. The **variable(s)**

1 The observational unit (a.k.a. **case**)



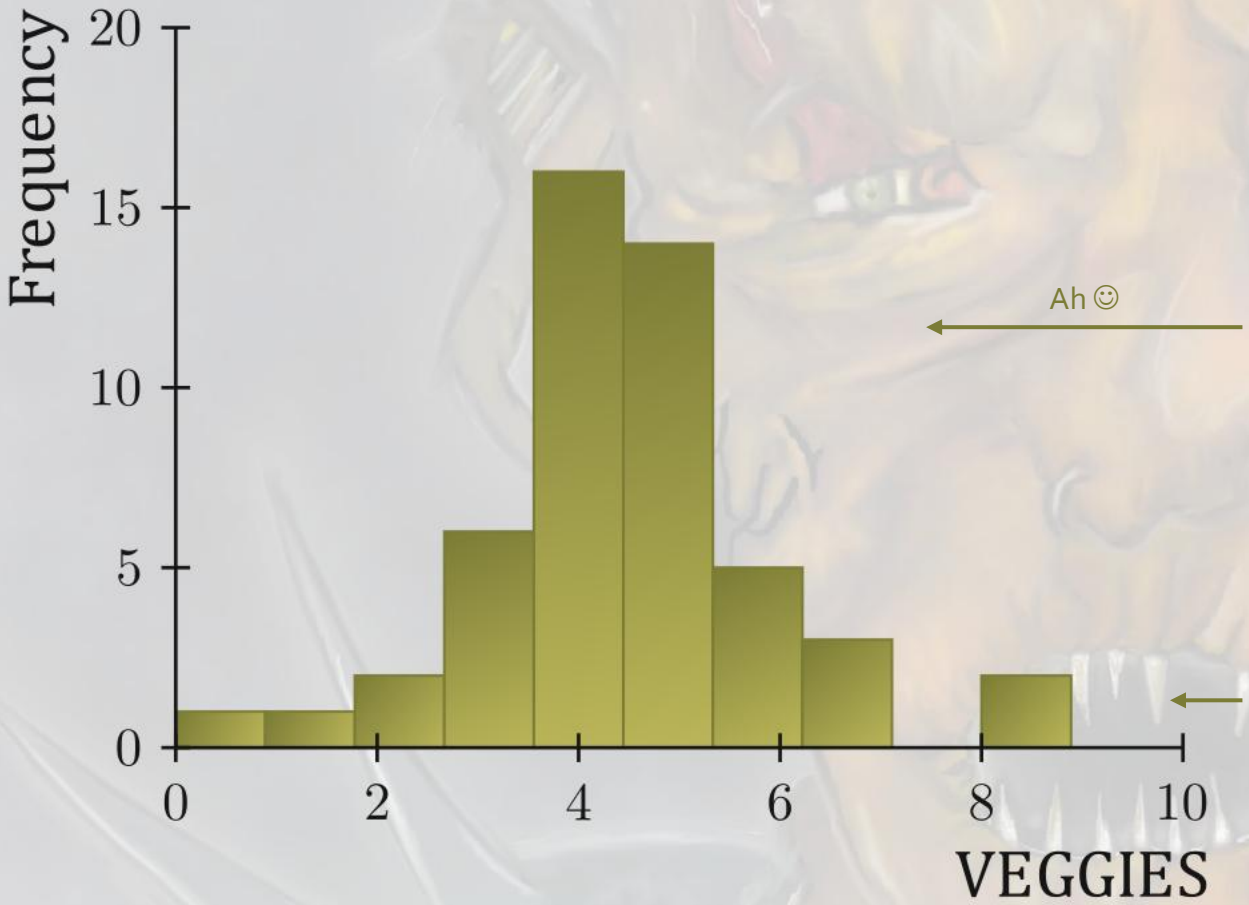
- Who, or what, is being measured/experimented upon/**observed**?
- 'A group of orcs?'
- Yes, but the **units** of that group are <u>individual soldiers</u>
  - So each individual orc is an observational unit
- 'Observational unit' is the One Term to Rule Them All. We can sometimes use a more specific term:
  - 'Participant' or 'subject'
    - in an experiment (psychological, social, or medical)
  - 'Respondent'
    - when you hand out surveys
  - Or just 'customer', 'company', 'record', …

2 The number of observations



- The Dark Lord does not spy on his entire army (even though he would have preferred it)
- Instead, he only takes a **sample** of orcs whom he observes
  - 50 orcs,
    so 50 observational units
  - We usually indicate this with the letter ***n***. As follows: $n = 50$

3 The variable(s)



- We now know *who* the Dark Lord observes, but *what characteristic or behaviour* does he observe?
- The answer: how many handfuls of veggies an orc eats over the course of one week
- This number will differ among the individual orcs. Therefore, it is a **variable**
- We can make fancy graphs about it, among other things
  - More on this particular graph (a histogram) next week

We have different **types of variables**

Categorical:

a bunch of different categories

Quantitative:

numerical quantities

Number of handfuls eaten

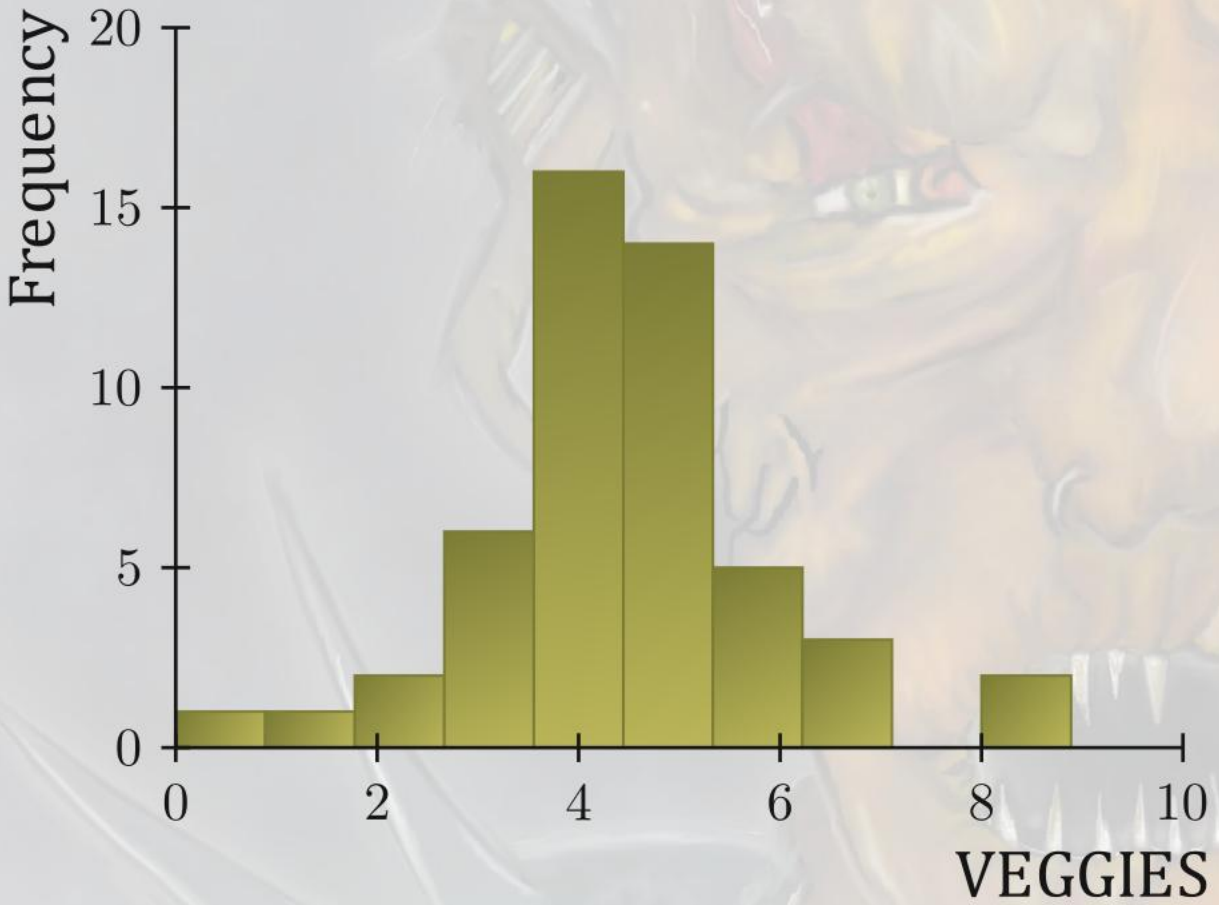
Type of vegetable
(carrot, beans, broccoli…)

How the food was prepared
(raw, steamed, stir-fried…)

The orc's rank
(soldier, captain, general…)

The orc's age in years

The variable type is crucial



It determines

- How you can summarise a variable's behaviour
- How you can visualise its behaviour
- What kinds of statistical tests you can unleash on it (weeks 5 to 7)

We should also distinguish the two subtypes of categorical variables

Nominal:

equal categories

Ordinal:

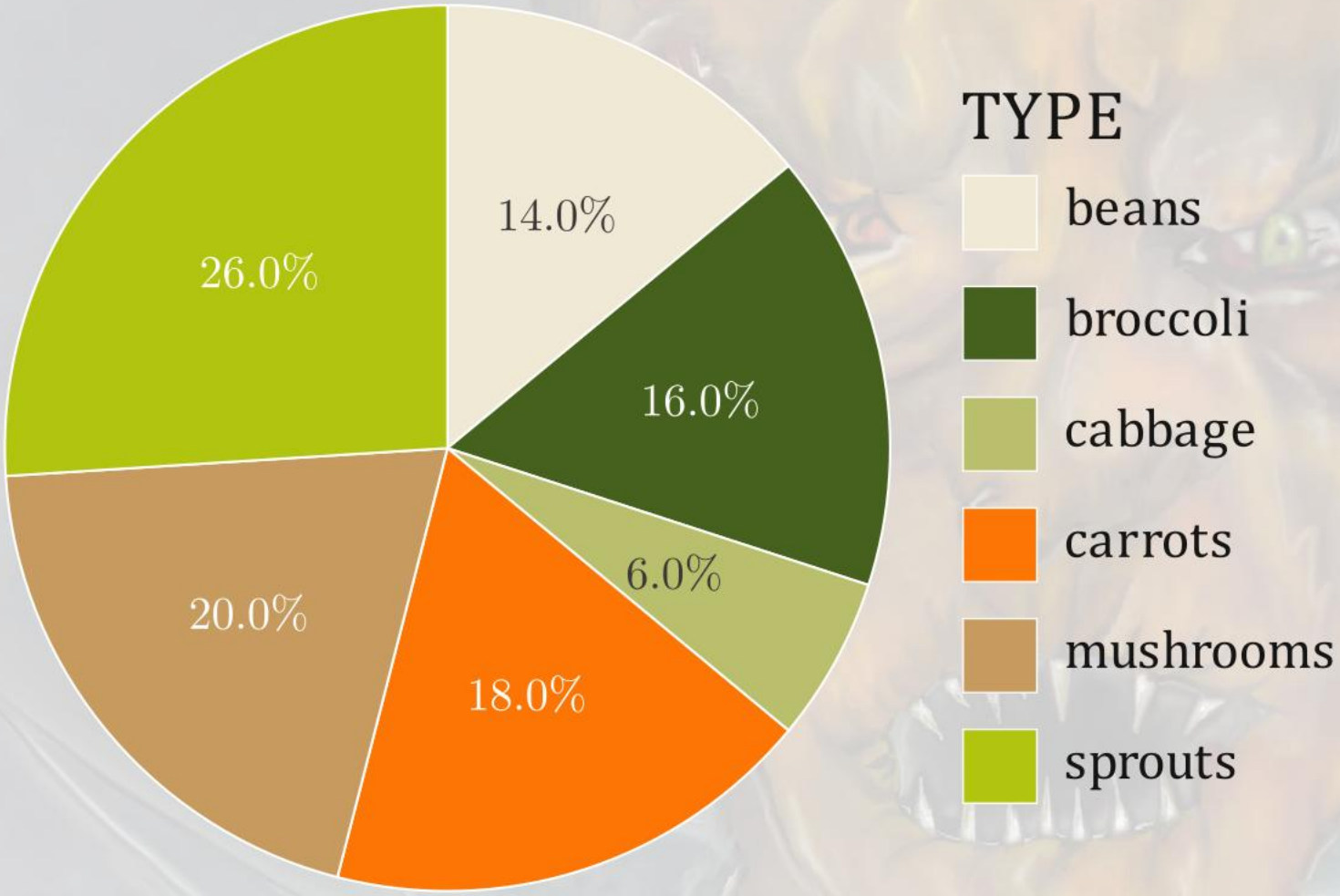
ordered categories

Type of vegetable
(carrot, beans, broccoli…)

How the food was prepared
(raw, steamed, stir-fried…)

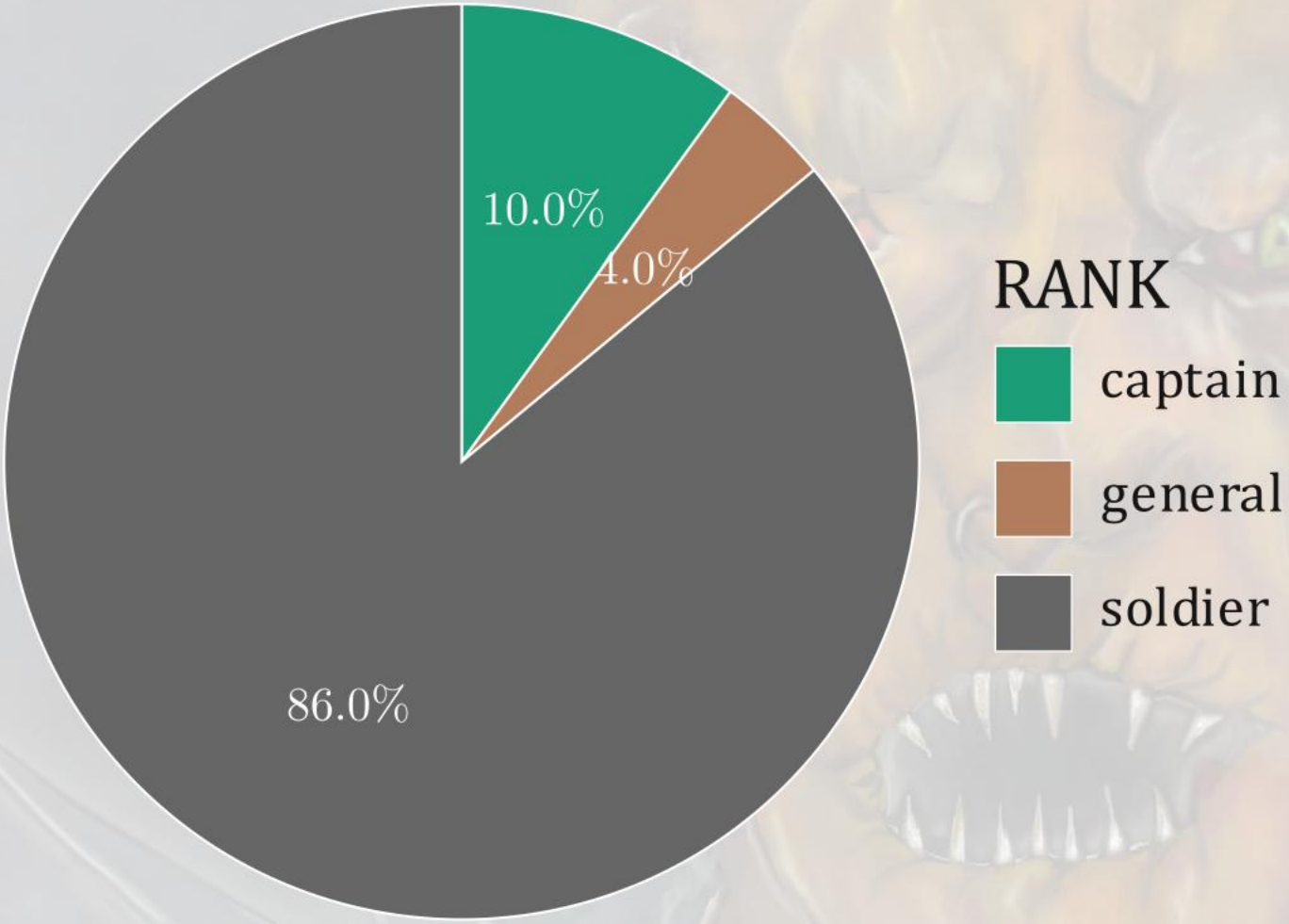
The orc's rank
(soldier, captain, general…)

How about some nice graphs?



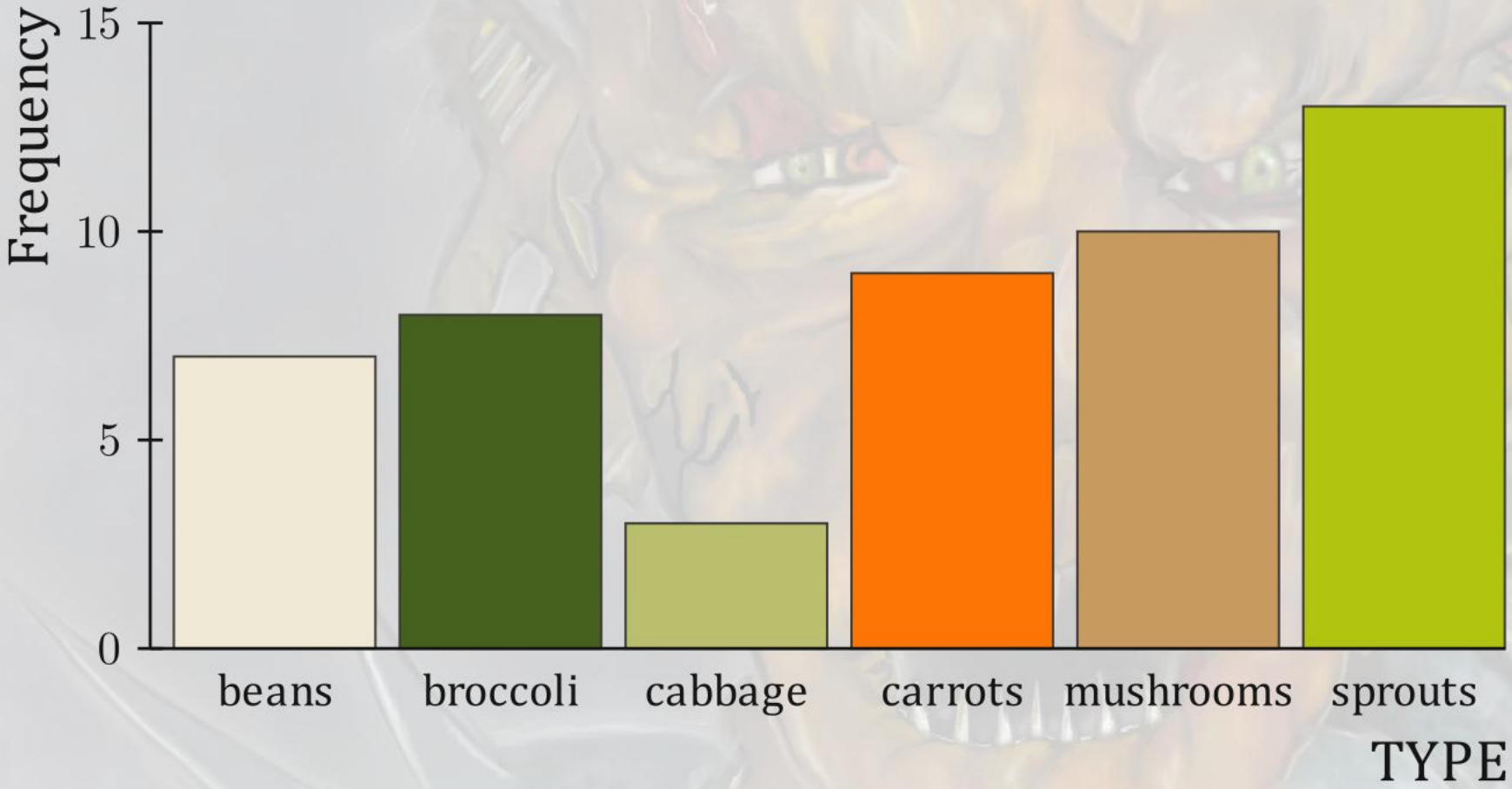
- Here's a **pie chart** of what vegetable the orcs ate on the first day
  - The entire pie represents the whole group (all 50 orcs)
  - About one quarter had sprouts, apparently
- Not an amazing type of chart
  - If you leave out the percentages, small differences become tough to see: more bean or more broccoli eaters?

How about some nice graphs? (continued)



- Another disadvantage: when your variable is <u>ordinal</u>, you cannot properly indicate the order
- Advice: use pie charts only to show how 1 or 2 categories stand out

How about some nice graphs? (continued)



- This **bar chart** indicates in another way how many orcs picked each veggie
- Much clearer ☺

- What to do with *quantitative* variables? We'll look into that next week

Tables work as well

PREPARATION	Frequency	Percent
raw	19	38
stewed	6	12
roasted	9	18
sautéed	7	14
steamed	9	18
Total	50	100

- A **frequency table** counts how many cases fall into each category
- It also compares these to the total number of observations, by showing percentages
  - Example: 19 out of 50 orcs ate their veggies raw, which amounts to

$$\frac{19}{50} * 100 \mathrel{\widehat{=}} 38\%$$

  - A pretty large amount, actually

Tables work as well (continued)

	TYPE						
PREPARATION	beans	broccoli	cabbage	carrots	mushrooms	sprouts	Total
raw	4	5	1	6	2	1	19
stewed	3	0	1	0	2	0	6
roasted	0	0	1	1	1	6	9
sautéed	0	2	0	2	3	0	7
steamed	0	1	0	0	2	6	9
Total	7	8	3	9	10	13	50

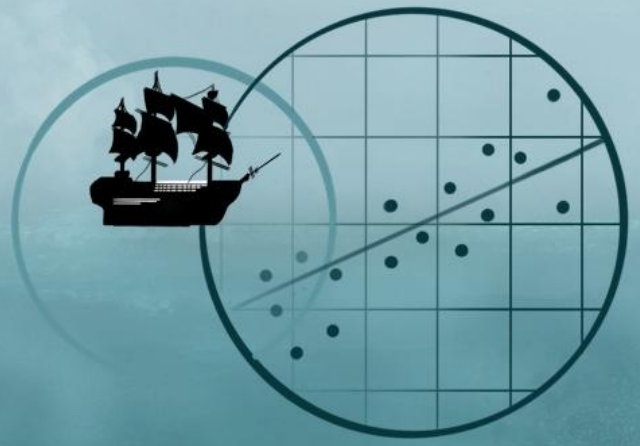
- A **contingency table** shows how <u>two</u> variables 'coincide'
  - In other languages, it goes by the name of 'cross tabulation': *kruistabel* (NL), *Kreuztabelle* (DE)
- Crossing two variables can show you interesting differences between groups, and/or relationships between variables
- We would expect a relationship here: some vegetables are nice to eat raw, while others should be cooked in some way
  - Nevertheless: most orcs who ate broccoli had it raw. Do they just not know what to do with it?
  - Perhaps the Dark Lord ought to spread some recipes

Further topics for this week (self-study ☺)

- Sharpe, chapter 1
  - The Five W's
- Sharpe, chapter 2
  - Other types of charts (e.g. segmented bar charts)
  - Simpson's paradox

- Keep in mind: we expect you to read the whole book chapter. This slide highlights the most important aspects that were not covered in this lecture

Take a break!

After this: mathematics

See you again soon! ☺